

MAT1002: Calculus II L01

Quiz 1 — 2025-2026, Term 2

Tuesday, 27 Jan 2026

1. (a) (5 points) Show that the sequence $\{a_n\}_{n=1}^{\infty}$ converges using the definition, where

$$a_n := \frac{n}{2n+1}.$$

Find its limit.

- (b) (5 points) Given that the sequence

$$\left\{ 2, 2 + \frac{1}{2}, 2 + \frac{1}{2 + \frac{1}{2}}, 2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}}, \dots \right\},$$

converges, find its limit.

Solution:

- (a) We compute

$$\left| a_n - \frac{1}{2} \right| = \left| \frac{n}{2n+1} - \frac{1}{2} \right| = \left| \frac{2n - (2n+1)}{2(2n+1)} \right| = \frac{1}{2(2n+1)}.$$

For all $n \geq 1$ we have $2(2n+1) = 4n+2 \geq 4n$, so

$$\left| a_n - \frac{1}{2} \right| \leq \frac{1}{4n}.$$

Given $\varepsilon > 0$, choose $N > \frac{1}{4\varepsilon}$. Then for all $n \geq N$,

$$\left| a_n - \frac{1}{2} \right| \leq \frac{1}{4n} \leq \frac{1}{4N} < \varepsilon.$$

Hence by the definition of limit $\lim_{n \rightarrow \infty} a_n = \frac{1}{2}$.

- (b) Let the terms be denoted $b_1 = 2$ and $b_{n+1} = 2 + \frac{1}{b_n}$ for $n \geq 1$. If the sequence converges to a limit L , then taking limits on both sides gives

$$L = 2 + \frac{1}{L}.$$

Multiply both sides by L :

$$L^2 = 2L + 1 \implies L^2 - 2L - 1 = 0.$$

Solving the quadratic,

$$L = \frac{2 \pm \sqrt{4+4}}{2} = 1 \pm \sqrt{2}.$$

Since every term of the sequence is positive, the limit must be nonnegative, so $L = 1 + \sqrt{2}$.

2. (a) (5 points) Discuss the convergence of the p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. You may use without proof any standard results.
- (b) (5 points) For $p = 2$, using the integral test approximate the series with $N = 9$. Find an upper bound for the error. (Note: $s_9 = 1 + \frac{1}{4} + \cdots + \frac{1}{9^2} \approx 1.53977$.)

Solution:

- (a) Consider the function $f(x) = x^{-p}$ on $[1, \infty)$. For $p \leq 0$ we have $a_n = 1/n^p$ does not tend to 0, so the series diverges. For $p > 0$, f is positive and eventually decreasing; the integral

$$\int_1^{\infty} x^{-p} dx$$

converges exactly when $p > 1$ and diverges for $p \leq 1$. By the integral test the p -series converges if and only if $p > 1$, and diverges for $p \leq 1$.

- (b) Let $S = \sum_{n=1}^{\infty} \frac{1}{n^2}$ and denote $s_N = \sum_{n=1}^N \frac{1}{n^2}$ and the remainder $R_N = S - s_N$. Since $f(x) = 1/x^2$ is positive and decreasing on $[1, \infty)$, the integral test inequalities give

$$\int_{N+1}^{\infty} \frac{dx}{x^2} \leq R_N \leq \int_N^{\infty} \frac{dx}{x^2}.$$

Evaluate the integrals:

$$\int_t^{\infty} \frac{dx}{x^2} = \left[-\frac{1}{x} \right]_t^{\infty} = \frac{1}{t}.$$

Hence

$$\frac{1}{N+1} \leq R_N \leq \frac{1}{N}.$$

For $N = 9$ we have

$$\frac{1}{10} = 0.1 \leq R_9 \leq \frac{1}{9} \approx 0.111111,$$

so

$$1.53977 + 0.1 \leq S \leq 1.53977 + 0.111111,$$

i.e.

$$1.63977 \leq S \leq 1.650881.$$

Therefore the error after nine terms satisfies $R_9 \leq \frac{1}{9} \approx 0.111111$. (A tighter statement is $0.1 \leq R_9 \leq 0.111111$.)

3. Justify whether the following series converge absolutely, converge conditionally, or diverge.

(a) (3 points) $\sum_{n=1}^{\infty} \frac{2^n + 5}{3^n}$.

(b) (3 points) $\sum_{n=1}^{\infty} a_n$, where

$$a_n = \begin{cases} \frac{n}{2^n}, & n \text{ is odd,} \\ \frac{1}{2^n}, & n \text{ is even.} \end{cases}$$

(c) (4 points) $\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{n^3 + 1}$.

Solution:

(a) Split the sum:

$$\sum_{n=1}^{\infty} \frac{2^n + 5}{3^n} = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n + 5 \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n.$$

Both are geometric series with ratios $2/3$ and $1/3$ respectively (both < 1), hence they converge absolutely. In fact,

$$\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \frac{2/3}{1 - 2/3} = 2, \quad 5 \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = 5 \cdot \frac{1/3}{1 - 1/3} = \frac{5}{2},$$

so the total sum equals $2 + \frac{5}{2} = \frac{9}{2}$.

(b) We show absolute convergence. Consider $|a_n|$. For even n we have $|a_n| = 1/2^n$, for odd n we have $|a_n| = n/2^n$. Use the root test (or compare with a convergent series):

$$\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \frac{1}{2},$$

because for odd n , $\sqrt[n]{n}/2 \rightarrow 1/2$, and for even n the n th root is $1/2$. Since $1/2 < 1$, the series $\sum |a_n|$ converges, hence $\sum a_n$ converges absolutely.

(c) Let $u_n = \frac{n^2}{n^3 + 1}$. First note $\lim_{n \rightarrow \infty} u_n = 0$ and for sufficiently large n the sequence (u_n) is positive and decreasing (one can check by differentiation of $f(x) = x^2/(x^3 + 1)$ or compare $u_{n+1} < u_n$ for large n). Therefore by the alternating series test, the alternating series $\sum (-1)^n u_n$ converges. For absolute convergence, compare

$$u_n \sim \frac{n^2}{n^3} = \frac{1}{n} \quad \text{as } n \rightarrow \infty,$$

so $\sum u_n$ behaves like the harmonic series and diverges. Thus the alternating series converges conditionally (convergent but not absolutely convergent).

MAT1002: Calculus II L02

Quiz 1 — 2025-2026, Term 2

For Tutors Only — Not For Distribution

Wednesday, 28 Jan 2026

1. (a) (6 points) Show that $\{(-1)^n\}_{n=1}^{\infty}$ diverges.

(b) (4 points) Find

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n}.$$

Solution:

(a) *Proof.* Suppose not. Then $\lim_{n \rightarrow \infty} (-1)^n = L$ for some $L \in \mathbb{R}$. Let $\epsilon = 1$. By definition, there exists N such that for all $n > N$, we have $|(-1)^n - L| < 1$. In particular, when n is even, we have $|L - 1| < 1$, i.e. $0 < L < 2$; if n is odd, we instead have $|L + 1| < 1$, i.e. $-2 < L < 0$. Such an L does not exist, which is a contradiction. Hence, $\{(-1)^n\}$ diverges. $\square$

(b)

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0,$$

where to get the 2nd equality, we have used L'Hôpital's rule.

2. Justify whether the following series converges. If so, calculate the limit.

(a) (2 points)

$$\sum_{n=1}^{\infty} \frac{e^n - 2}{4e^n + 5}.$$

(b) (2 points)

$$\sum_{n=1}^{\infty} \sin \frac{n\pi}{2}.$$

(c) (3 points)

$$\sum_{n=1}^{\infty} \frac{1}{n}.$$

(d) (3 points)

$$\sum_{n=3}^{\infty} \sin \frac{1}{n}.$$

Solution:

(a) The series diverges since

$$\lim_{n \rightarrow \infty} \frac{e^n - 2}{4e^n + 5} = \frac{1}{4} \neq 0.$$

(b) The series diverges since

$$\lim_{n \rightarrow \infty} \sin \frac{n\pi}{2}$$

does not exist.

(c) Consider $f(x) = 1/x$, which is positive and decreasing on $[1, \infty)$. Since

$$\int_1^{\infty} \frac{1}{x} dx$$

diverges, and by the integral test, we know that the harmonic series diverges.

(d) Compare with $\sum 1/n$. Note that

$$\lim_{n \rightarrow \infty} \frac{\sin \frac{1}{n}}{\frac{1}{n}} = \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

And from the last part, the harmonic series diverges. So the series

$$\sum_{n=3}^{\infty} \sin \frac{1}{n}$$

diverges.

3. Justify whether the following series converges absolutely or conditionally, or diverges.

(a) (3 points)

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}.$$

(b) (3 points)

$$\sum_{n=1}^{\infty} \frac{2^n}{n^3}.$$

(c) (4 points)

$$\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}.$$

Solution:

(a) Since

$$\lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \lim_{n \rightarrow \infty} \frac{(n+1)n^n}{(n+1)(n+1)^n} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{1}{e} < 1.$$

By Ratio Test, the series converges absolutely.

(b) Note that

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \frac{2}{(\sqrt[n]{n})^3} = 2 > 1.$$

By Root Test, the series diverges.

(c) Note that the sequence $\{1/n\}$ is positive and decreasing on $[1, \infty]$ and $1/n \rightarrow 0$ as $n \rightarrow \infty$. By the alternating series test, it converges. Further, since $\sum 1/n$ diverges. The series converges conditionally.

MAT1002: Calculus II L01

Quiz 2 — 2025-2026, Term 2

Tuesday, 3 Mar 2026

1. (a) (6 points) Express the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1},$$

in the form of an elementary function. Identify its domain.

- (b) (10 points) Show that the Maclaurin series of $f(x) := e^x$ converges to $f(x)$ for all $x \in \mathbb{R}$.

Solution:

- (a) By Ratio Test,

$$\left| \frac{a_{n+1}}{a_n} \right| = \left(\frac{2n+1}{2n+3} \right) |x|^2 \rightarrow |x|^2,$$

as $n \rightarrow \infty$. The series converges when $|x| < 1$ and diverges for $|x| > 1$. Let the function f be defined as $f(x) := \sum (-1)^n x^{2n+1} / (2n+1)$ for $-1 < x < 1$. Consider its derivative

$$f'(x) = \sum_{n=0}^{\infty} (-1)x^{2n} = \sum_{n=0}^{\infty} (-x^2)^n = \frac{1}{1+x^2},$$

for $|x| < 1$. Then

$$f(x) = \int f'(x) dx = \int \frac{1}{1+x^2} dx = \arctan x + C,$$

where C is a constant. Using the fact that $0 = f(0) = \arctan 0 + C = C$, we know $f(x) = \arctan x, x \in (-1, 1)$. Note that it is okay to include $x = 1$ (No penalty if students do not include endpoints).

- (b) The function $f(x) = e^x$ has derivatives of all orders for $x \in \mathbb{R}$. Note that

$$f^{(k)}(x) = e^x \implies f^{(k)}(0) = 1, k \in \mathbb{N}.$$

By Taylor's theorem, we write

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^k}{k!} + R_k(x),$$

where the remainder

$$R_k(x) = \frac{e^c}{(k+1)!} x^{k+1},$$

for some c between 0 and x . Since e^x is an increasing function, so e^c lies between 1 and e^x . If $x \leq 0$, then $e^x < e^c < 1$ and

$$|R_k(x)| \leq \frac{|x|^{k+1}}{(k+1)!}.$$

If $x > 0$, then $1 < e^c < e^x$ and

$$|R_k(x)| \leq e^x \frac{|x|^{k+1}}{(k+1)!}.$$

Finally, because $x^{k+1}/(k+1)! \rightarrow 0$ as $k \rightarrow \infty$ for every x ,

$$\lim_{k \rightarrow \infty} R_k(x) = 0,$$

and the series converges to e^x for every x . Thus,

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

2. (4 points) Find the limit

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right).$$

Solution: We have

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) &= \lim_{x \rightarrow 0} \frac{x - \sin x}{x \sin x} \\ &= \lim_{x \rightarrow 0} \frac{x - (x + \mathcal{O}(x^3))}{x(x + \mathcal{O}(x^3))} \\ &= \lim_{x \rightarrow 0} \frac{\mathcal{O}(x^3)}{x^2 + \mathcal{O}(x^4)} \\ &= \lim_{x \rightarrow 0} \frac{\mathcal{O}(x)}{1 + \mathcal{O}(x^2)} \\ &= \frac{0}{1 + 0} \\ &= 0. \end{aligned}$$

3. (a) (5 points) Calculate area of the surface generated by revolving about the x -axis the curve given by $x = \cos t, y = 1 + \sin t, t \in [0, 2\pi]$.
- (b) (5 points) Find the area under one arch of the cycloid $x = r(\theta - \sin \theta), y = r(1 - \cos \theta), \theta \in [0, 2\pi]$.

Solution:

- (a) For $0 \leq t \leq 2\pi$, the area is

$$S = \int_0^{2\pi} 2\pi y(t) \sqrt{\left(\frac{dy}{dt}\right)^2 + \left(\frac{dx}{dt}\right)^2} dt = 2\pi \int_0^{2\pi} (1 + \sin t) \cdot 1 dt = (2\pi)^2.$$

- (b) The area is

$$\begin{aligned} A &= \int_0^{2\pi} y(\theta) \frac{dx(\theta)}{d\theta} d\theta \\ &= \int_0^{2\pi} r(1 - \cos \theta) r(1 - \cos \theta) d\theta \\ &= r^2 \int_0^{2\pi} (1 - 2\cos \theta + \cos^2 \theta) d\theta \\ &= r^2 \left(2\pi + \frac{\cos 2\theta + 1}{2} \Big|_{\theta=0}^{2\pi} \right) \\ &= 3\pi r^2. \end{aligned}$$

MAT1002: Calculus II L02

Quiz 2 — 2025-2026, Term 2

Wednesday, 4 Mar 2026

1. (a) (5 points) Express the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1},$$

in the form of an elementary function. Identify its domain.

- (b) (10 points) Show that the Maclaurin series of $f(x) := \sin x$ converges to $f(x)$ for all $x \in \mathbb{R}$.

Solution:

- (a) By Ratio Test,

$$\left| \frac{a_{n+1}}{a_n} \right| = \left(\frac{n+1}{n+2} \right) |x| \rightarrow |x|,$$

as $n \rightarrow \infty$. The series converges when $|x| < 1$ and diverges for $|x| > 1$. Let the function f be defined as $f(x) := \sum (-1)^n x^{n+1} / (n+1)$ for $-1 < x < 1$. Consider its derivative

$$f'(x) = \sum_{n=0}^{\infty} (-1)^n x^n = \sum_{n=0}^{\infty} (-x)^n = \frac{1}{1+x},$$

for $|x| < 1$. Then

$$f(x) = \int f'(x) dx = \int \frac{1}{1+x} dx = \ln |1+x| + C,$$

where C is a constant. Using the fact that $0 = f(0) = \ln 1 + C = C$, we have $f(x) = \ln(1+x)$, $x \in (-1, 1)$. Note that it is okay to include $x = 1$ (No penalty if students do not include endpoints).

- (b) The function $f(x) = \sin x$ has derivatives of all orders for $x \in \mathbb{R}$. Note that

$$f'(x) = \cos(x), f''(x) = -\sin x, f'''(x) = -\cos x \implies f^{(2k)}(0) = 0, f^{(2k+1)}(0) = (-1)^k, k \in \mathbb{N}.$$

By Taylor's theorem, we write

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^k x^{2k+1}}{(2k+1)!} + R_{2k+1}(x),$$

where $R_{2k+1}(x)$ is a remainder. Since $|\cos(x)| \leq 1$, we have

$$|R_{2k+1}(x)| \leq 1 \cdot \frac{|x|^{2k+2}}{(2k+2)!} \rightarrow 0$$

as $n \rightarrow \infty$ and the series converges to $\sin x$ for every x . Thus,

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

2. (5 points) Find the limit

$$\lim_{x \rightarrow 0} \frac{\cos x - 1 + \frac{1}{2}x \sin x}{[\ln(1+x)]^4}.$$

Solution: We have

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\cos x - 1 + \frac{1}{2}x \sin x}{[\ln(1+x)]^4} \\ &= \lim_{x \rightarrow 0} \frac{1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \mathcal{O}(x^6) - 1 + \frac{x}{2} \left(x - \frac{x^3}{3!} + \mathcal{O}(x^5) \right)}{[x + \mathcal{O}(x^2)]^4} \\ &= \lim_{x \rightarrow 0} \frac{\frac{x^4}{4!} - \frac{x^4}{3! \cdot 2} + \frac{x}{2} \mathcal{O}(x^5) + \mathcal{O}(x^6)}{[x + \mathcal{O}(x^2)]^4} \\ &= \lim_{x \rightarrow 0} \frac{\left(\frac{1}{4!} - \frac{1}{3! \cdot 2} \right) + \mathcal{O}(x^2)}{[1 + \mathcal{O}(x)]^4} \\ &= \frac{1}{4!} - \frac{1}{3! \cdot 2} \\ &= \frac{1}{24} - \frac{1}{12} \\ &= -\frac{1}{24}. \end{aligned}$$

3. (a) (5 points) Consider the plane curve given by $x = t^2, y = t^3 - 3t, t \in \mathbb{R}$. Express the 2nd order derivative d^2y/dx^2 in terms of t .
- (b) (5 points) Find the length of the parametric curve given by $x = \sin(2t), y = \cos(2t), t \in [0, 2\pi]$.

Solution:

- (a) We have

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d(dy/dx)/dt}{dx/dt},$$

and

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3(t-1)}{2t}.$$

Substituting dy/dx back into the 1st equation, we have

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{3t-3}{2t} \right)}{2t} = \frac{3(t^2+1)}{4t^3}.$$

- (b) The curve is the unit circle, but traced twice. It is traced exactly once when t goes from 0 to π . So the length is

$$L = \int_0^\pi \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^\pi \sqrt{4\cos^2(2t) + 4\sin^2(2t)} dt = 2 \int_0^\pi dt = 2\pi.$$

MAT1002: Calculus II L01

Quiz 3 — 2025-2026, Term 2

Tuesday, 7 Apr 2026

1. (a) (5 points) Show that the following limit does not exist,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}.$$

- (b) (5 points) Does

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2},$$

exist? If so, prove it; if not, justify your answer.

Solution:

- (a) Along the curve $x = ky^2$, $(x, y) \rightarrow 0$ as $y \rightarrow 0$, so

$$\lim_{(x,y) \rightarrow (0,0) \text{ along } x=ky^2} \frac{xy^2}{x^2 + y^4} = \lim_{y \rightarrow 0} \frac{ky^4}{(k^2 + 1)y^4} = \frac{k}{k^2 + 1} = \begin{cases} \frac{1}{2}, & k = 1, \\ -\frac{1}{2}, & k = -1. \end{cases}$$

Hence, the limit does not exist.

- (b) For arbitrary $\epsilon > 0$, we let $\delta = \epsilon/3$, then for all (x, y) with $0 < \sqrt{x^2 + y^2} < \delta$, we have

$$\left| \frac{3x^2y}{x^2 + y^2} \right| < 3 \left| \frac{x^2y}{x^2} \right| < 3|y| \leq 3\sqrt{x^2 + y^2} < 3\delta = \epsilon,$$

so

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2} = 0.$$

Students may use other approaches to prove it.

2. (a) (4 points) Suppose that $z = e^x \sin y$, $x = st^2$ and $y = s^2t$. Find $\partial z / \partial s$ and $\partial z / \partial t$.
(b) (3 points) Find the slope of the tangent line to the curve $x^3 + y^3 = 6xy$ at the point $(3, 3)$.
(c) (3 points) Find the directional derivative of $f(x, y) := xe^y + \cos(xy)$ at the point $(2, 0)$ in the direction of $\vec{v} = \langle 3, -4 \rangle$.

Solution:

- (a)

$$\frac{\partial z}{\partial s} = t^2 \sin(s^2t) e^{st^2} + 2st \cos(s^2t) e^{st^2}.$$

$$\frac{\partial z}{\partial t} = 2st \sin(s^2 t) e^{st^2} + s^2 \cos(s^2 t) e^{st^2}.$$

(b) Let $F(x, y) = x^3 + y^3 - 6xy$, so $F_x = 3x^2 - 6y$ and $F_y = 3y^2 - 6x$. Then

$$\left. \frac{dy}{dx} \right|_{(x,y)=(3,5)} = -\frac{F_x(3, 3)}{F_y(3, 3)} = -1.$$

(c) The unit vector along $\vec{v}$ is $\hat{v} = \langle 3/5, -4/5 \rangle$. The directional derivative of f at $(2, 0)$ in the direction of $\hat{v}$ is

$$\begin{aligned} D_{\vec{v}} f(2, 0) &= \lim_{h \rightarrow 0} \frac{f(2 + 3h/5, -4h/5) - f(2, 0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2 + 3h/5)e^{-4h/5} + \cos[(2 + 3h/5)(-4h/5)] - (2 + 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2(e^{-4h/5} - 1)}{h} + \lim_{h \rightarrow 0} \frac{3}{5} e^{-4h/5} + \lim_{h \rightarrow 0} \frac{\cos(-8h/5 - 12h^2/5) - 1}{h} \\ &= 2 \left. \frac{d[e^{-4h/5}]}{dh} \right|_{h \rightarrow 0} + \frac{3}{5} + 2 \left. \frac{d[\cos(-8h/5 - 12h^2/5)]}{dh} \right|_{h \rightarrow 0} \\ &= -\frac{8}{5} + \frac{3}{5} \\ &= -1. \end{aligned}$$

Alternatively, we can directly calculate

$$D_{\hat{v}} f(2, 0) = \nabla f(2, 0) \cdot \hat{v} = \langle 1, 2 \rangle \cdot \left\langle \frac{3}{5}, -\frac{4}{5} \right\rangle = -1.$$

3. (a) (6 points) For $f(x, y, z) = x^2 - xy + 3 \sin z$, consider the standard linear approximation at $P_0(2, 1, 0)$ on

$$R = [1.99, 2.01] \times [0.98, 1.02] \times [-0.01, 0.01].$$

Find the linear approximation from P_0 in R and estimate the error.

- (b) (4 points) Find all critical points of the function

$$f(x, y) := x^4 + y^4 - 4xy + 1,$$

defined on $\mathbb{R}^2$.

Solution:

- (a) At P_0 , we have

$$f(P_0) = 2, f_x(P_0) = 3, f_y(P_0) = -2, f_z(P_0) = 3.$$

The linear approximation is

$$L(x, y, z) = 2 + 3(x - 2) - 2(y - 1) + 3z = 3x - 2y + 3z - 2.$$

Since $f_{xx} = 2$, $f_{yy} = 0$, $f_{zz} = -3 \sin z$, $f_{xy} = -1$, $f_{xz} = 0$, $f_{yz} = 0$, all their absolute values are ≤ 2 . Hence, for all (x, y, z) in R ,

$$|E(x, y, z)| \leq \frac{1}{2} \cdot 2(|x - 2| + |y - 1| + |z - 0|)^2 \leq (0.01 + 0.02 + 0.01)^2 = 0.0016.$$

(b) Solving

$$f_x = 4x^3 - 4y = 0, f_y = 4y^3 - 4x = 0,$$

we have three solutions (critical points): $(x, y) = (1, 1)$ or $(x, y) = (-1, -1)$ or $(x, y) = (0, 0)$.

MAT1002: Calculus II L02

Quiz 3 — 2025-2026, Term 2

Wednesday, 8 Apr 2026

1. (a) (5 points) Show that the following limit does not exist,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}.$$

- (b) (5 points) Does

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2},$$

exist? If so, prove it; if not, justify it.

Solution:

- (a) Along the curve $y = kx$, $(x, y) \rightarrow 0$ as $y \rightarrow 0$, so

$$\lim_{(x,y) \rightarrow (0,0) \text{ along } y=kx} \frac{xy}{x^2 + y^2} = \lim_{y \rightarrow 0} \frac{xkx}{x^2 + (kx)^2} = \frac{k}{k^2 + 1} = \begin{cases} \frac{1}{2}, & \text{along } y = x, \\ -\frac{1}{2}, & \text{along } y = -x. \end{cases}$$

Hence, the limit does not exist.

- (b) On the x -axis, i.e., $y = 0, x \neq 0$, $f(x, y) = (x^2 - y^2)/(x^2 + y^2) \equiv 1$. On the y -axis, i.e., $x = 0, y \neq 0$, $f(x, y) \equiv -1$. Hence,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

does not exist.

2. (a) (6 points) Let $\hat{u} = \langle u_1, u_2 \rangle$ be any unit vector, find the directional derivative of $f(x, y)$ at $(0, 0)$ in the direction of $\hat{u}$ with

$$f(x, y) := \begin{cases} \frac{xy^2}{x^2 + y^4}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Is $f(x, y)$ differentiable at $(0, 0)$?

- (b) (4 points) Find an equation for the tangent line to the ellipse $x^2/8 + y^2/2 = 1$ at the point $(-2, 1)$.

Solution:

(a) We have

$$\begin{aligned} D_{\vec{u}}f(0,0) &= \lim_{h \rightarrow 0} \frac{f(hu_1, hu_2) - f(0,0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^3 u_1 u_2^2}{h(h^2 u_1^2 + h^4 u_2^4)} \\ &= \lim_{h \rightarrow 0} \frac{u_1 u_2^2}{u_1^2 + h^2 u_2^4} \\ &= \begin{cases} \frac{u_2^2}{u_1}, & u_1 \neq 0, \\ 0, & u_1 = 0. \end{cases} \end{aligned}$$

The two variable $f(x, y)$ is not differentiable at $(0, 0)$.

(b) Let $f(x, y) = x^2/8 + y^2/2$, then $f_x = x/4$ and $f_y = y$. Therefore, $f_x(-2, 1) = -1/2$ and $f_y(-2, 1) = 1$. The tangent line is

$$-\frac{1}{2}(x+2) + (y-1) = 0,$$

or

$$y = x/2 + 2.$$

3. (10 points) Find the global maximum and global minimum of the function

$$f(x, y) := 2 + 2x + 4y - x^2 - y^2$$

defined on the triangular region in the first quadrant bounded by the lines $x = 0, y = 0$ and $9 - x$.

Solution: We have $f_x = 2 - 2x, f_y = 4 - 2y$. Solving $f_x = f_y = 0$ gives $(x, y) = (1, 2)$, which is in the interior of $\mathbb{D}$. The boundary is $l_1 \cup l_2 \cup l_3$, with

$$l_1 := \{(0, y) : 0 \leq y \leq 9\}; \quad l_2 := \{(x, 0) : 0 \leq x \leq 9\}; \quad l_3 := \{(x, 9 - x) : 0 \leq x \leq 9\}.$$

On l_3 , $f(x, y) = f(x, 9 - x) = -43 + 16x - 2x^2 = p(x)$. Solving $p'(x) = 0$ gives $x = 4$, so $(x, y) = (4, 5)$ is a candidate. At end points $x = 0$ and $x = 9$, we have $(0, 9)$ and $(9, 0)$ also being candidates. By doing similar arithmetic for l_1 and l_2 , we can find the following candidates:

$$l_1 : (0, 0), (0, 2), (0, 9); \quad l_2 : (0, 0), (1, 0), (9, 0).$$

Computing the values of f at all these candidates,

$$f(1, 2) = 7, f(0, 0) = 2, f(0, 2) = 6, f(0, 9) = -43, f(1, 0) = 3, f(9, 0) = -61, f(4, 5) = -11,$$

we find that the global maximum is $f(1, 2) = 7$ and the global minimum is $f(9, 0) = -61$.

MAT1002: Calculus II L01

Quiz 4 — 2025-2026, Term 2

Tuesday, 21 Apr 2026

1. (a) (4 points) Evaluate the double integral $\iint_D (x + 2y) dA$, where D is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.
(b) (6 points) Evaluate $\iiint_E z dV$, where E is the solid tetrahedron bounded by the planes $x = 0, y = 0, z = 0$ and $x + y + z = 1$.
2. (a) (3 points) A solid E lies inside the cylinder $x^2 + y^2 = 1$, below the plane $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$. The density at a point $P(x, y)$ is $K\sqrt{x^2 + y^2}$ for some constant K . Find $\text{mass}(E)$.
(b) (7 points) Find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$.
3. (10 points) Evaluate the integral

$$\iint_D e^{\frac{x+y}{x-y}} dA,$$

where D is the trapezoidal region with vertices $(1, 0), (2, 0), (0, 1)$ and $(0, 2)$.

MAT1002: Calculus II L02

Quiz 4 — 2025-2026, Term 2

Wednesday, 22 Apr 2026

1. (a) (3 points) Evaluate the iterated integral $\int_0^1 \int_x^1 \sin(y^2) dy dx$.
- (b) (7 points) Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, above the xy -plane, and inside the cylinder $x^2 + y^2 = 2x$.

Solution:

- (a) Since $\sin(y^2)$ is continuous, we have

$$\int_0^1 \int_x^1 \sin(y^2) dy dx = \int_0^1 \int_0^y \sin(y^2) dx dy = \int_0^1 y \sin(y^2) dy = \frac{1 - \cos 1}{2}.$$

- (b) Note that $x^2 + y^2 = 2x \Leftrightarrow (x - 1)^2 + y^2 = 1$, which is a circle, say D , centered at $(1, 0)$ with radius 1. The volume is

$$V = \iint_D \int_0^{x^2+y^2} dz dA.$$

Let $x = 1 + r \cos \theta$, $y = r \sin \theta$, $r \geq 0$, $\theta \in [0, 2\pi)$. So $x^2 + y^2 = r^2 + 2r \cos \theta + 1$, and

$$V = \int_0^{2\pi} \int_0^1 (r^2 + 2r \cos \theta + 1) r dr d\theta = \int_0^{2\pi} \left(\frac{1}{4} + \frac{2}{3} \cos \theta + \frac{1}{2} \right) d\theta = \frac{3\pi}{2}.$$

2. (a) (4 points) Use cylindrical coordinates to evaluate the integral

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2 + y^2) dz dy dx.$$

- (b) (6 points) Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{3/2}} dV$, where B is the closed ball centered at the origin with radius 1.

Solution:

- (a) In cylindrical coordinates (r, θ, z) , $x = r \cos \theta$, $y = r \sin \theta$, $z = z$ with $r \geq 0$, $\theta \in$

$[0, 2\pi)$. We have

$$\begin{aligned} \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) dz dy dx &= \int_{\theta=0}^{2\pi} \int_{r=0}^2 \int_{z=r}^2 r^2 dz r dr d\theta \\ &= 2\pi \int_{r=0}^2 r^3 (2-r) dr \\ &= 2\pi \left(\frac{16}{2} - \frac{32}{5} \right) \\ &= \frac{16\pi}{5}. \end{aligned}$$

(b) In spherical polar coordinates (r, θ, ϕ) , $x = r \cos \phi \sin \theta$, $y = r \sin \phi \sin \theta$, $z = r \cos \theta$ with $r \geq 0$, $\phi \in [0, 2\pi)$, $\theta \in [0, \pi]$. We have

$$\iiint_B e^{(x^2+y^2+z^2)^{3/2}} dV = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^1 r^2 \sin \theta e^{r^3} dr d\theta d\phi = \frac{4\pi(e-1)}{3}.$$

3. (10 points) Evaluate the integral

$$\int_1^2 \int_{1/y}^y \sqrt{\frac{y}{x}} e^{\sqrt{xy}} dx dy.$$

Solution: Let $u = \sqrt{xy}$; $v = \sqrt{y/x}$. Then $x = u/v$; $y = uv$, with $u, v > 0$. The Jacobian is

$$\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{2u}{v} > 0.$$

The boundary of D in (x, y) -coordinates consists of the following lines:

$$y = x; \quad y = -x; \quad y = 2.$$

In terms of (u, v) , they map to

$$v = 1; \quad v = -1; \quad uv = 2.$$

Thus, after the change of variables, we have

$$\int_{u=1}^2 \int_{v=1}^{2/u} 2ue^u dv du = 2 \int_1^2 e^u (2-u) du = 2e(e-2).$$